SPACE MISSION

ANALYTICAL DASHBOARDS

End-to-End Data Pipeline · MySQL · Power BI · 5-Page Dashboard

1,584	7	68	333	47
Missions	Countries	Years	Rockets	Launch Sites

Dataset: 1,584 Records | 1957–2025 | 23 Columns | 36→23 Cleaned | 7 SQL Views

PROJECT INTRODUCTION

What is This Project About?

🎯 **Objective**	Analyse 68 years of global space missions to uncover patterns in launch activity, mission success rates, rocket reliability, and orbital strategy across 7 major space programs.
🛠️ **Tools Used**	Python (data cleaning) · MySQL (import, type conversion, SQL views) · Power BI Desktop (5-page interactive dashboard)
📦 **Dataset**	1,584 missions spanning 1957–2025. Original 36-column raw CSV cleaned to 23 final columns with 2 new derived fields (Decade, Spacecraft_Status).
🌍 **Scope**	7 country groups: USA, Russia/USSR, China, India, Europe/ESA, Japan, Other Countries. Covers all mission types from Military/Recon to Human Spaceflight.
📊 **Deliverable**	Interactive 6-page Power BI report (Home + 5 analysis pages) with 7 SQL views, 7 DAX measures, and synced global slicers.

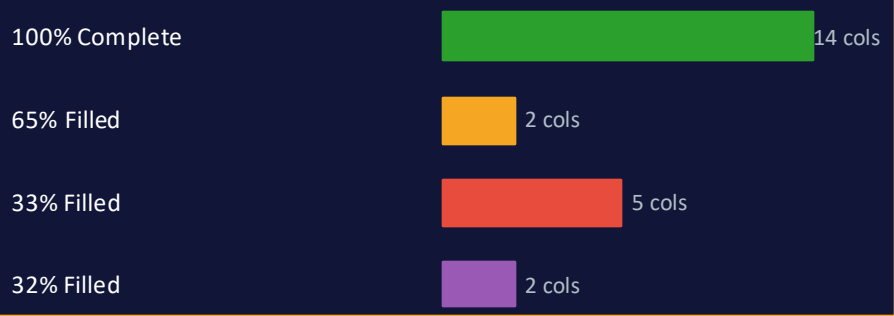
DATASET OVERVIEW

23 Final Columns — By Group

IDENTITY (4)	Mission_Name · Launch_Date · Launch_Year · Decade★
GEOGRAPHY (3)	Country_Group · Country · Agency
VEHICLE (4)	Launch_Vehicle · Rocket_Family · Launch_Site · Launch_Complex
MISSION (2)	Mission_Type (12 cats) · Orbit_Class (13 cats)
PHYSICAL (1)	Satellite_Mass_kg (32% filled — 512 rows)
OUTCOME (2)	Mission_Outcome (4 groups) · Spacecraft_Status★
ECONOMICS (3)	Estimated_Cost_USD_M · Fuel_Type · Operational_Life_Yrs
METADATA (4)	Notes · Launch_ID · Data_Source · Dataset_Source

★ = newly derived column

Column Fill Rates



Cleaning Summary

- 36 raw columns → 23 final columns
- 13 columns dropped (redundant / unusable)
- 7 columns normalised (types, categories)
- 2 new columns derived (Decade, Spacecraft_Status)
- 7 problem date rows fixed manually
- Empty strings converted to NULL (launch_site)
- Cost ranges converted to numeric midpoints

DATA CLEANING PROCESS

1 — Raw Import

Create missions_raw table with ALL columns as VARCHAR — guarantees zero import errors on any data format.

2 — CSV Import

Import MASTER_CLEANED_FINAL.csv via MySQL Workbench Table Data Import Wizard into missions_raw.

3 — Fix 7 Dates

Manually fix 7 problem rows (range dates, planned dates, mixed formats) by mission name using MAKEDATE().

4 — Type Convert

Add clean columns, convert VARCHAR→DATE, SMALLINT, DECIMAL using STR_TO_DATE, CAST, NULLIF, TRIM.

5 — Build Clean Table

CREATE missions_clean with proper data types. INSERT SELECT from missions_raw using converted columns.

Critical Concept — NULL vs Empty String vs 'Unknown'

State	Meaning	Correct SQL Filter	Example Column
NULL	No value — blank cell in CSV	WHERE col IS NOT NULL	satellite_mass_kg
Empty String ""	Cell existed with no content	WHERE col != ""	fuel_type
"Unknown"	Real text value = Unknown	WHERE col NOT IN ("Unknown")	orbit_class

DATA UNDERSTANDING — KEY FIELDS & STRUCTURE

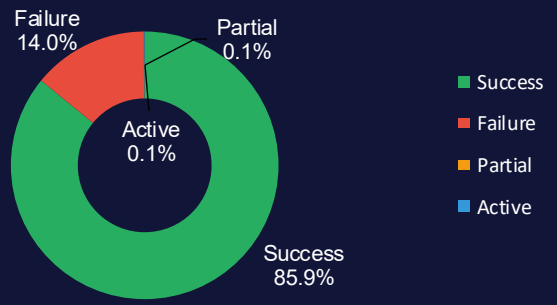
Important Fields Explained

Country_Group	7 clean groups — primary slicer for all dashboards. Never use raw Country column in visuals.
Mission_Outcome	4 values: Success (86%), Failure (14%), Partial, Active. BepiColombo = Active (launched 2018).
Orbit_Class	13 types. Has 'Unknown' and 'Other' text — filter: NOT IN ('Unknown','Other').
Fuel_Type	5 families. Missing stored as empty string '' not NULL — filter: fuel_type != ''.
Decade	Derived column: '1960s','1970s'... for decade-level grouping. 100% filled.
Spacecraft_Status	Derived: Decommissioned / Active / Lost-Failed / Planned. Only 33% filled.
Estimated_Cost_USD_M	Numeric midpoints from ranges. Only 503/1584 rows (32%) — always add coverage note.

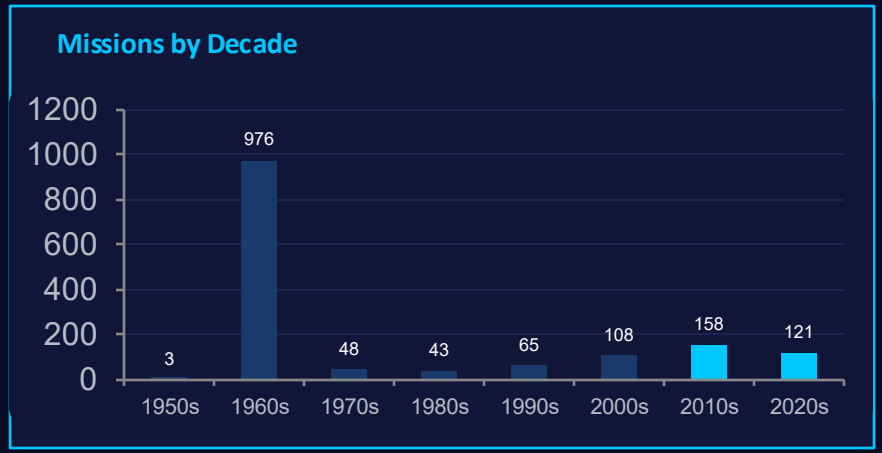
Data Pipeline

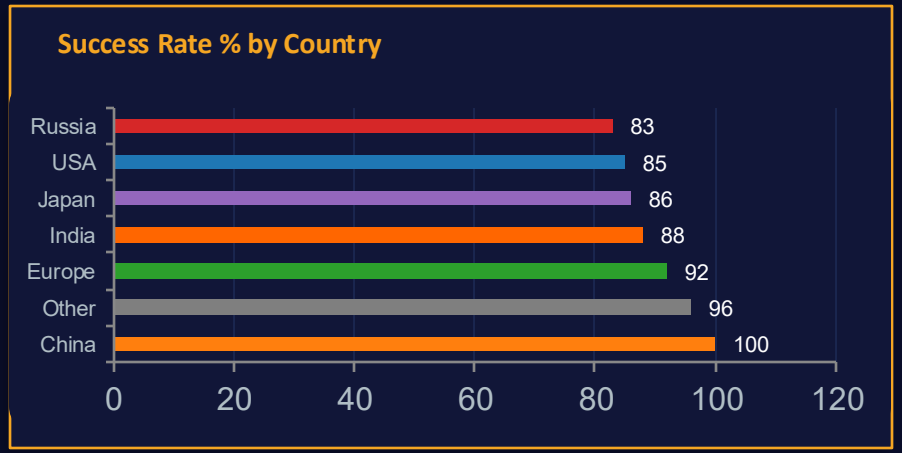
1. Raw CSV (36 cols)
2. Python Cleaning
3. 23-col CSV
4. MySQL Raw Table
5. Type Conversions
6. missions_clean
7. 7 SQL Views
8. **Power BI Dashboard**

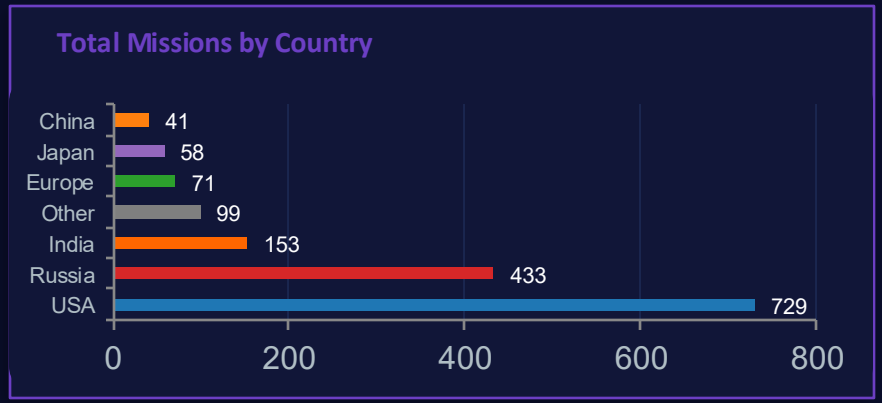
Mission Outcome Split



EXPLORATORY DATA ANALYSIS — KEY INSIGHTS







SQL VIEWS CREATION — 7 VIEWS FOR POWER BI

vw_mission_summary

Core flat table — all 23 key columns. Used by most visuals across all 5 pages.

Filter: launch_year <= 2025, NULLIF(launch_site, '')

vw_orbit_analysis

Orbital distribution — mission count, avg mass, success rate per orbit class.

Filter: orbit_class NOT IN ('Unknown','Other')

vw_country_performance

Country KPIs — total missions, success rate, avg cost, type diversity per country group.

Filter: launch_year <= 2025

vw_launch_site_map

Geographic data — 47 launch sites with coordinates, success rate, first/last launch year.

Filter: launch_site IS NOT NULL AND != ''

vw_yearly_trends

Time series — missions and success rate by year, decade, and country for trend charts.

Filter: launch_year IS NOT NULL AND <= 2025

vw_technology_evolution

Fuel & propulsion trends — missions and avg cost by decade, fuel type, mission type.

Filter: fuel_type != '' AND decade IS NOT NULL

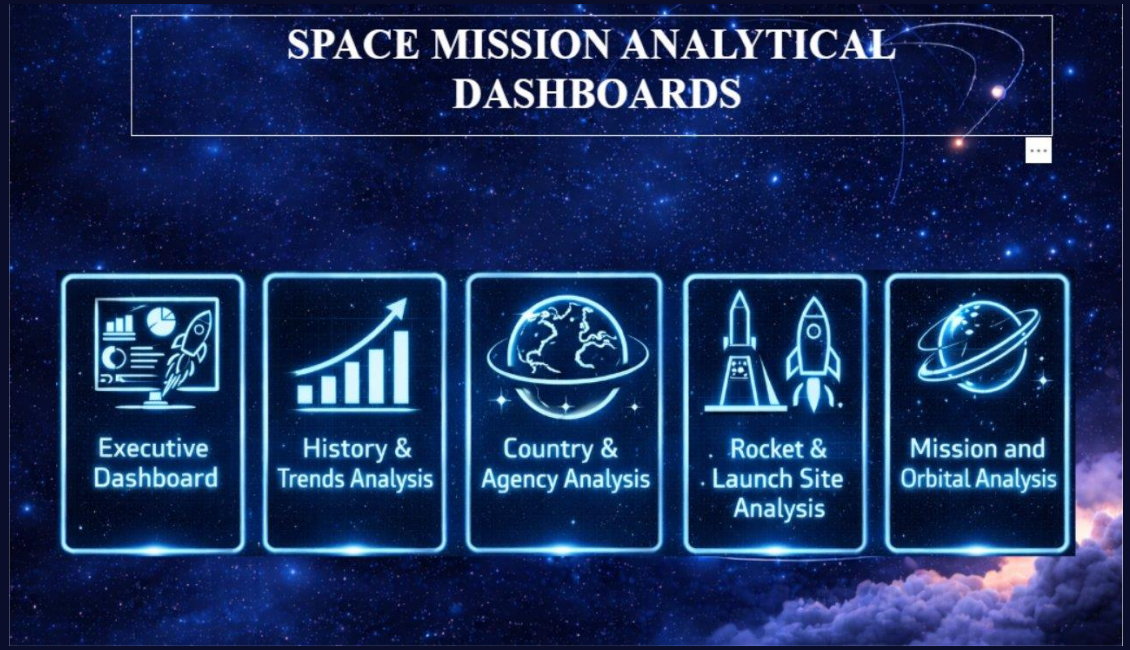
vw_rocket_reliability

Vehicle analysis — reliability %, total launches, payload avg. HAVING total_launches >= 3.

Filter: launch_year <= 2025, min 3 launches

Key Principle — Each view uses the correct WHERE filter based on how that column stores missing data (NULL vs '' vs 'Unknown'). Wrong filter = wrong numbers.

DASHBOARD DEVELOPMENT — 5-PAGE DESIGN



6-Page Report Structure

Home	Navigation hub with 5 dashboard buttons
Page 1	Executive Overview — KPIs + trends
Page 2	History & Trends — decade analysis
Page 3	Country & Agency — deep dive
Page 4	Rockets & Launch Sites
Page 5	Mission & Orbital Analysis

Global Slicers — Synced Across All 5 Pages

country_group · launch_year (range) · decade (tiles) · mission_outcome · mission_type

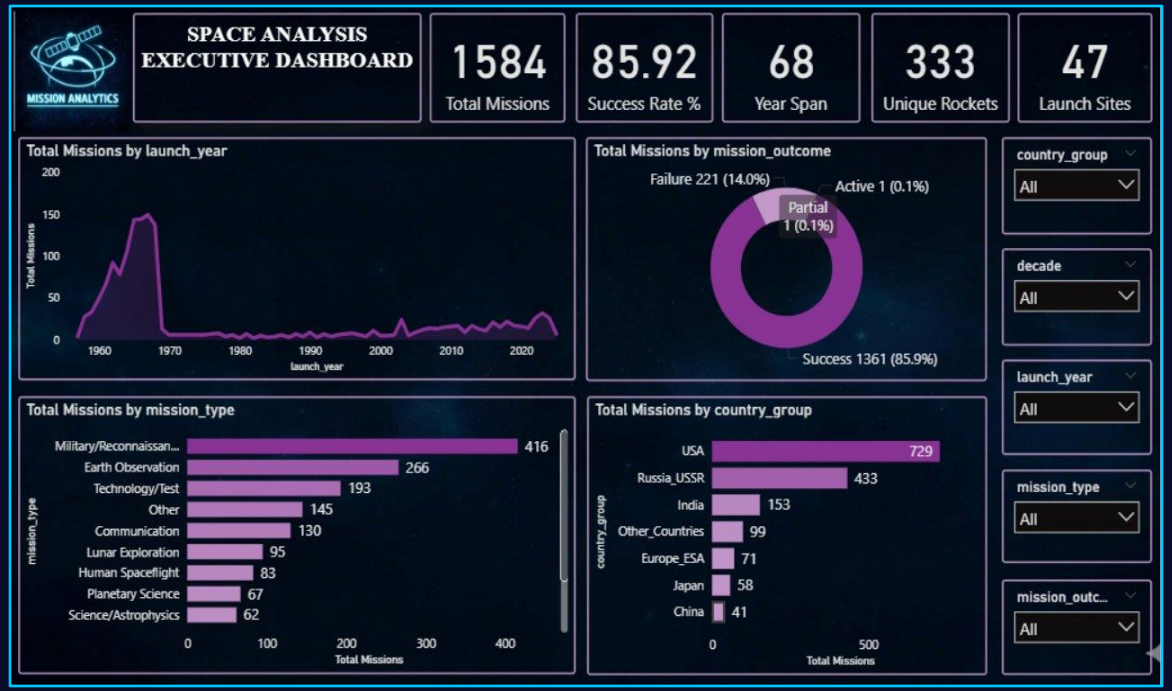
Enable sync: View → Sync Slicers → tick all 5 pages for each slicer

7 DAX Measures

Total Missions · Success Rate % · Failure Rate %
Avg Mass kg · Avg Cost USD M
Missions With Cost · Cumulative Missions

DASHBOARD 1 — EXECUTIVE OVERVIEW

1,584	85.9%	68 Yrs	333
Total Missions	Success Rate	Year Span	Unique Rockets



Key Insights

📌 USA Leads

729 missions — 46% of all launches. Followed by Russia/USSR with 433 (27%).

📉 Space Race Peak

1960s account for 976 missions — more than all other decades combined.

✅ High Success

1,361 successes vs 221 failures. Overall success rate of 85.9%.

🛰 Mission Mix

Military/Recon (26%) and Earth Observation (17%) dominate mission types.

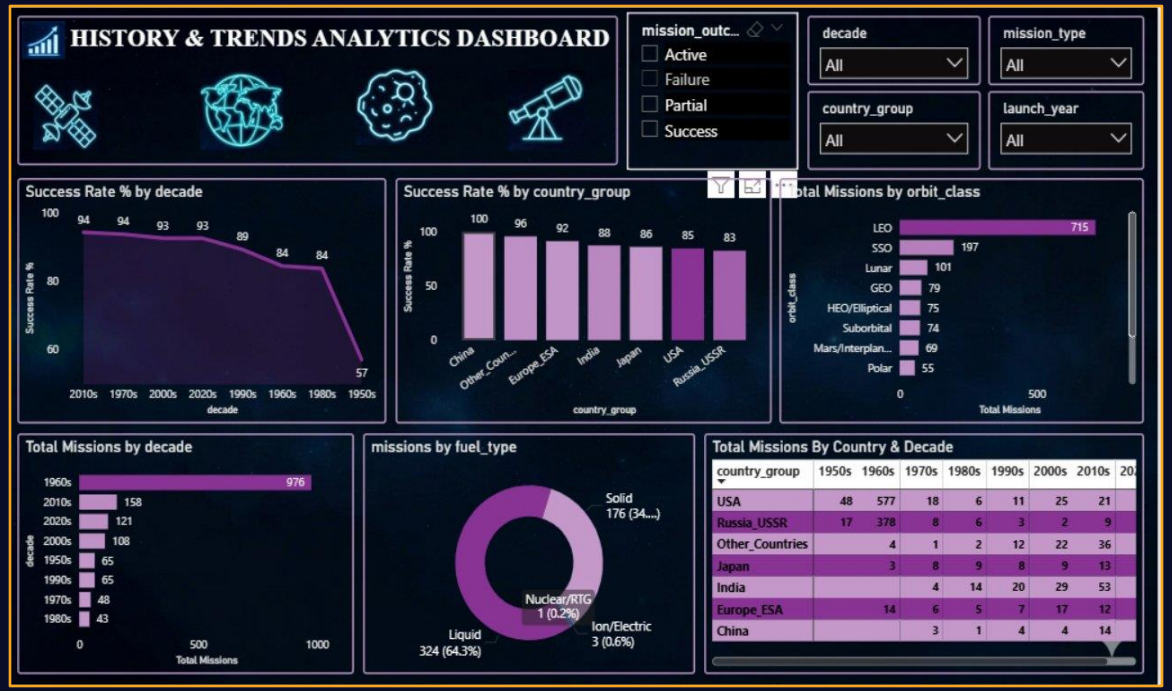
DASHBOARD 2 — HISTORY & TRENDS ANALYSIS

57%
1950s Success Rate

94%
2010s Success Rate

976
1960s Missions

64%
Liquid Fuel Share



Key Insights

Dramatic Improvement

Success rate rose from 57% in the 1950s to 94% in the 2010s — a 37-point gain.

Cold War Spike

1960s Space Race drove an unprecedented volume peak — 62% of all recorded missions.

Fuel Shift

Liquid propellant dominates at 64%. Ion/Electric engines rising sharply from 2010s onward.

Country Heatmap

Russia/USSR was absent post-2010s in the matrix — replaced by China's rapid expansion.

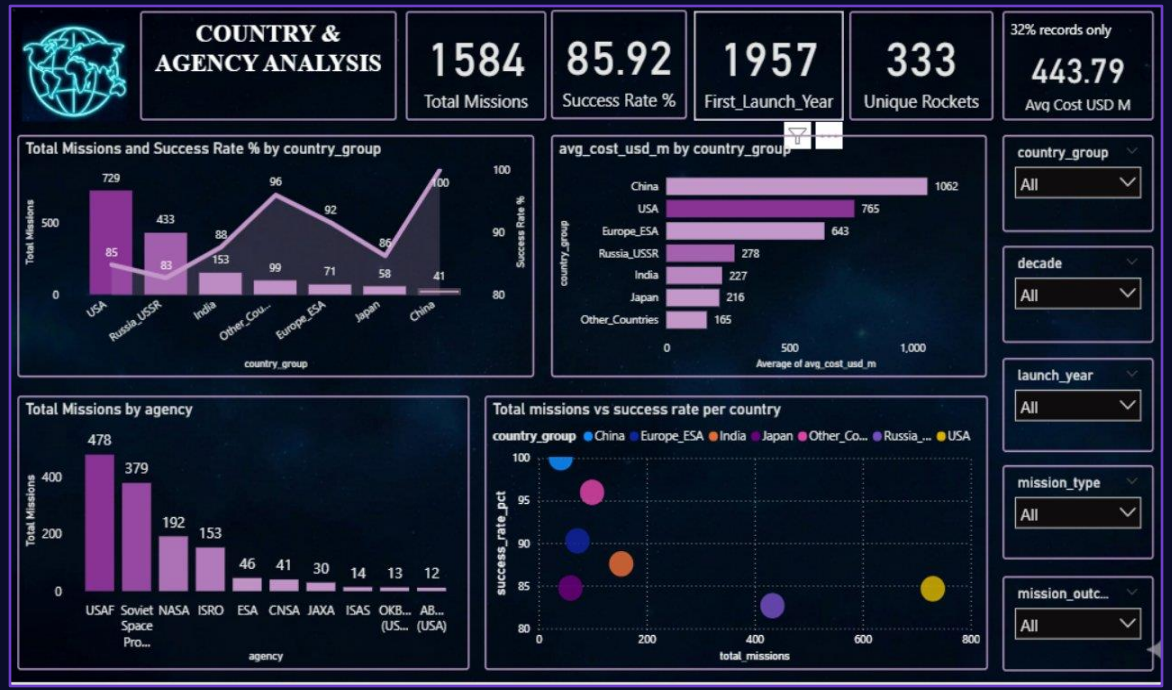
DASHBOARD 3 — COUNTRY & AGENCY ANALYSIS

100%
China Success Rate

$1,062M
China Avg Cost (highest)

$227M
India Avg Cost (lowest)

478
USAF Missions (top agency)



Key Insights

China Perfection

China achieved a 100% success rate across all 41 recorded missions — most remarkable record.

ISRO Rising

India's ISRO shows the steepest modern growth — 88% success on significantly lower budgets.

Cost Leaders

China avg $1,062M is highest; India $227M is lowest — over 4x difference in cost efficiency.

Agency Dominance

USAF (478) and Soviet Space Program (379) collectively account for 54% of all missions.

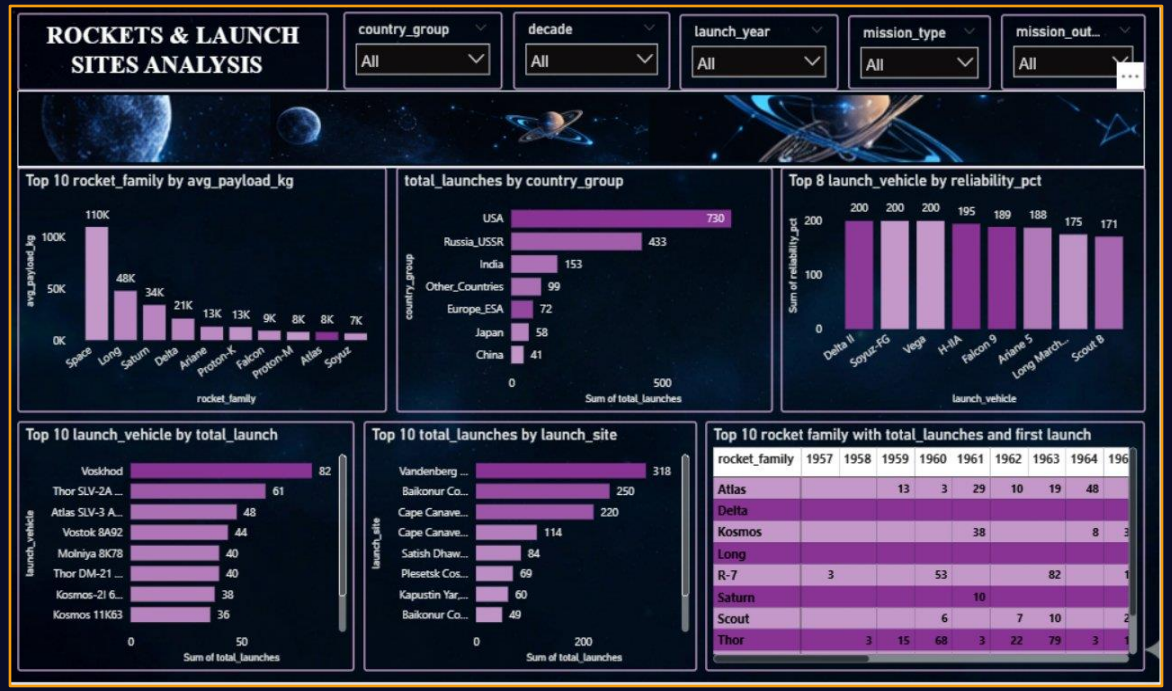
DASHBOARD 4 — ROCKETS & LAUNCH SITES

82
Voskhod Launches (most used)

318
Vandenberg Launches (top site)

95%+
Delta II / Soyuz-FG Reliability

333
Unique Rocket Variants



Key Insights

🚀 Most Flown

Voskhod leads with 82 launches. Top 5 rockets are all Cold War-era US/Soviet vehicles.

📍 Top 3 Sites

Vandenberg (318), Baikonur (250) and Cape Canaveral (220) — 3 sites = 50% of all launches.

🛡️ High Reliability

Delta II, Soyuz-FG, Vega and H-IIA all exceed 95% reliability — proven veteran platforms.

⚡ Falcon 9 Era

The matrix shows Falcon 9 emergence in 2010s — representing the commercial reusable revolution.

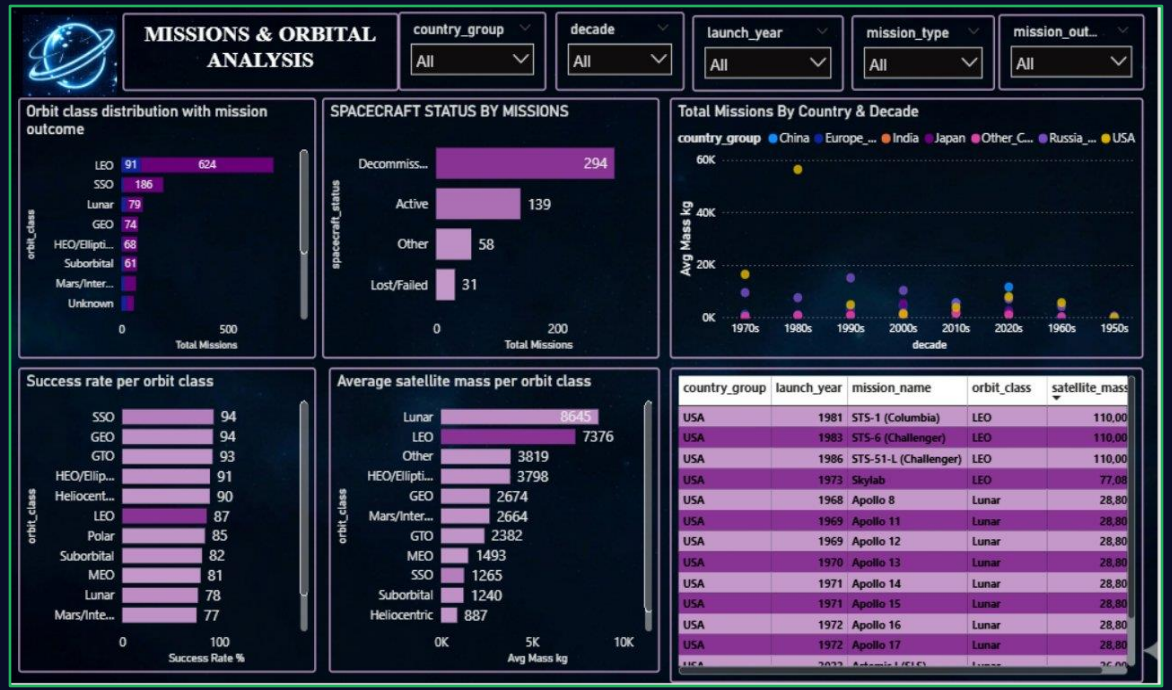
DASHBOARD 5 — MISSION & ORBITAL ANALYSIS

715
LEO Missions (45% of total)

94%
SSO Success Rate (highest)

8,645 kg
Avg Lunar Payload (heaviest)

77%
Mars Success Rate (lowest)



Key Insights

LEO Dominates

Low Earth Orbit hosts 715 missions (45%) — driven by ISS, Earth observation, and cubesats.

SSO Precision

Sun-synchronous orbit has the highest success rate at 94% — well-understood and repeatable.

Lunar Heavy

Lunar missions carry the heaviest average payloads at 8,645 kg due to propulsion demands.

Mars is Hard

Mars/Interplanetary has the lowest success at 77% — deep-space navigation remains the frontier.

INSIGHTS, CONCLUSION & RECOMMENDATIONS

Key Insights

🏆 **Success Rate**	China achieved 100% success rate. Russia/USSR lowest at 83%, reflecting early Cold War failures.
📈 **Space Race Peak**	1960s had 976 missions — 62% of all missions. Decade accounts for biggest spike in launch activity.
🚀 **Most Launched**	Voskhod (82 launches) is the most used rocket. USA and Russia dominate top 10 vehicles.
🌍 **Launch Sites**	Vandenberg (318), Baikonur (250) and Cape Canaveral (220) are the top 3 sites globally.
🛸 **Orbit Preference**	LEO dominates at 715 missions (45%). SSO, Lunar and GEO follow as next most used orbits.
💰 **Cost Data**	Only 32% of missions have cost data. China avg $1,062M is highest, India lowest at $227M.
⚡ **Fuel Evolution**	Liquid fuel dominates (64%). Ion/Electric rising in 2010s-2020s for commercial satellites.

Conclusion

- Successfully cleaned 36-column raw dataset to 23 production-ready columns.
- MySQL import pipeline with proper type handling eliminates all data errors.
- 7 SQL views provide clean, filtered data optimised for each dashboard page.
- 5-page Power BI dashboard delivers actionable insights with global slicer sync.
- 88% of missions are fully verified via Launch Library 2 UUID traceability.

Recommendations & Future Scope

- Fill cost data gaps using public sources to increase 32% coverage.
- Add launch coordinates for all 47 sites to enable accurate map visuals.
- Automate monthly data refresh by connecting to Launch Library 2 API.
- Add crew manifest data to enhance Human Spaceflight mission analysis.
- Build ML model to predict mission success probability from vehicle + orbit.
- Expand to include payload operator and customer for commercial analysis.